Spectral Methods for Variational Quantum Linear Solvers

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Outline

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3. Main Results
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Why Should Anyone Care?

The big picture

- Many scientific problems reduce to solving a *linear system* $A\mathbf{x} = \mathbf{b}$
- Classical algorithms cost $O(N^3)$ (Gaussian elimination) or $O(N^2)$ (iterative) — expensive for large N
- **HHL quantum algorithm**: $O(\log N)$ depth — exponential speedup in N

NISQ era reality

Current quantum hardware is noisy. **Variational Quantum Linear Solver (VQLS)** works on near-term devices, but its cost depends critically on how many *Pauli terms* appear in A.

Central question

Can we *choose* how to discretize a differential equation so that the resulting matrix has **fewer Pauli terms**, making VQLS faster?

Pauli Strings — A Quick Reminder

Single-qubit Paulis:

$$I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

Multi-qubit Pauli strings are tensor products, e.g. $X \otimes Z \otimes I$ for 3 qubits.

Key facts

- $\{I, X, Y, Z\}^{\otimes n}$ is an orthogonal basis for all $2^n \times 2^n$ matrices
- For n qubits: 4^n Pauli strings total
- $n = 2$: up to 16 strings; $n = 3$: up to 64
- Coefficient: $c_P = \frac{1}{2^n} \operatorname{Tr}(A\,P)$

Why fewer Paulis helps

Each nonzero c_P costs circuit evaluations in VQLS. A sparser Pauli decomposition means fewer measurements, less quantum overhead.

Collocation Discretization — Simple Example

Problem: $u'' = 0$, $u(0) = 0$, $u(1) = 1$. *Exact solution:* $u(t) = t$.

Spectral collocation ($N = 4$, monomial basis $\{1, t, t^2, t^3\}$):

1. Nodes: boundaries $t = 0, 1$ and interior nodes $t = \frac{1}{3}, \frac{2}{3}$
2. Write $u(t) \approx c_0 + c_1 t + c_2 t^2 + c_3 t^3$
3. Enforce BCs at rows 0–1; enforce $u'' = 0$ at rows 2–3

$$\underbrace{\begin{pmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 \\ 0 & 0 & 2 & 2 \\ 0 & 0 & 2 & 4 \end{pmatrix}}_{A_{\text{mono}}} \begin{pmatrix} c_0 \\ c_1 \\ c_2 \\ c_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}$$

A_{mono} uses **all 16 Pauli strings** ($4^2 = 16$ for 2 qubits). Can we change basis to get fewer?

Experiment 1 — Results

Applying the four conditions to the trial function $u(t) = \sum_{i=0}^{3} a_i \phi_i(t)$ yields the system:

$$\begin{bmatrix} C_{00} & C_{10} & C_{20} & C_{30} \\ C_{00} & (C_{11}+C_{10}) & \sum C_{2j} & \sum C_{3j} \\ 0 & 0 & 2C_{22} & (2C_{33}+2C_{32}) \\ 0 & 0 & 2C_{22} & (4C_{33}+2C_{32}) \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ a_2 \\ a_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix} \tag{4}$$

The determinant of this matrix is given by:

$$4C_{00}C_{11}C_{22}C_{33} \tag{5}$$

4. Minimizing the Number of Pauli Strings

The Pauli decomposition of any 4×4 matrix A is unique: the coefficient of each two-qubit Pauli string P_k is determined by

$$c_k = \frac{\mathrm{Tr}(P_k A)}{4}. \tag{6}$$

There are 16 such strings in total. The goal is to choose the basis coefficients C_{ij} so that as many c_k as possible are exactly zero. Three structural conditions, imposed in sequence, achieve this.

Step 1: Enforce block-diagonal structure. The upper-right 2×2 block of the general Case B matrix has entries

$$\begin{pmatrix} C_{20} & C_{30} \\ C_{20}+C_{21}+C_{22} & C_{30}+C_{31}+C_{32}+C_{33} \end{pmatrix}. \tag{7}$$

This block is nonzero for generic choices of C_{ij} (e.g. $C_{30} = 1$ immediately places a nonzero entry at position $(0, 3)$). To force it to zero we impose:

$$C_{20} = 0, \qquad C_{30} = 0, \qquad C_{21} + C_{22} = 0, \qquad C_{31} + C_{32} + C_{33} = 0. \tag{8}$$

The lower-left block is already zero by construction (the collocation rows evaluate ϕ_0'' and ϕ_1'', which vanish at interior points). With the above constraints, the matrix takes the block form

$$A = \begin{pmatrix} B_1 & 0 \\ 0 & B_2 \end{pmatrix}, \qquad B_1 = \begin{bmatrix} C_{00} & C_{10} \\ C_{00} & C_{11}+C_{10} \end{bmatrix}, \quad B_2 = \begin{bmatrix} 2C_{22} & 2C_{33}+2C_{32} \\ 2C_{22} & 4C_{33}+2C_{32} \end{bmatrix}. \tag{9}$$

Step 3: Repeated first-column entry (eliminates IY). For a 2×2 matrix $B = \left[\begin{smallmatrix} a & b \\ a & c \end{smallmatrix}\right]$, the coefficient of Y in its single-qubit Pauli decomposition is $\mathrm{Tr}(YB)/2 = i(b-a)/2$. This vanishes if and only if $b = a$, i.e. the off-diagonal entry equals the top-left entry. For B_1 this requires $C_{10} = C_{00}$, and since Step 2 already forces $B_1 = B_2$, the same condition holds for B_2 automatically. After imposing $C_{10} = C_{00}$, only II, IX, and IZ remain.

Why three is the minimum. With the block $B = \left[\begin{smallmatrix} a & a \\ a & c \end{smallmatrix}\right]$, the three surviving coefficients are

$$\alpha = \frac{a+c}{2}, \qquad \beta = a, \qquad \gamma = \frac{a-c}{2}, \tag{11}$$

so that $A = \alpha\, II + \beta\, IX + \gamma\, IZ$ (before normalisation). Eliminating IZ would require $a = c$, making $\det(B) = a(c-a) = 0$; the system would have no unique solution. Hence three Pauli strings is the minimum compatible with a non-singular collocation matrix.

5. Resulting Decomposition

Applying all three conditions with the concrete choice

$$\begin{aligned} &C_{00} = 2, \\ &C_{10} = 2, \quad C_{11} = 2, \\ &C_{20} = 0, \quad C_{21} = -1, \quad C_{22} = 1, \\ &C_{30} = 0, \quad C_{31} = -1, \quad C_{32} = 0, \quad C_{33} = 1 \end{aligned}$$

gives $a = 2$, $c = 4$, and the matrix

$$A = \begin{bmatrix} 2 & 2 & 0 & 0 \\ 2 & 4 & 0 & 0 \\ 0 & 0 & 2 & 2 \\ 0 & 0 & 2 & 4 \end{bmatrix}. \tag{12}$$

After normalising by $\|A\| = \sqrt{6a^2 + 2c^2} = 2\sqrt{14}$, the Pauli decomposition is

$$\frac{A}{\|A\|} = \frac{3}{2\sqrt{14}}\, II + \frac{1}{\sqrt{14}}\, IX - \frac{1}{2\sqrt{14}}\, IZ \approx 0.4009\, II + 0.2673\, IX - 0.1336\, IZ, \tag{13}$$

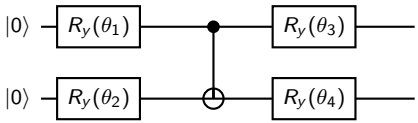
which is an 81% reduction from the 16 Pauli strings required in Case A.

Variational Quantum Linear Solver (VQLS)

Goal: solve $A|x\rangle = |b\rangle$ on a NISQ device.

Algorithm loop:

1. Prepare ansatz $|\psi(\boldsymbol{\theta})\rangle = V(\boldsymbol{\theta})|0\rangle$
2. Evaluate cost $C(\boldsymbol{\theta}) = 1 - \dfrac{|\langle b|A|\psi\rangle|^2}{\langle\psi|A^\dagger A|\psi\rangle}$
3. Update $\boldsymbol{\theta}$ by gradient descent
4. Repeat until $C \approx 0$



Hardware-efficient ansatz (2 qubits)

Analogy: supervised ML

ML	VQLS
Neural net	Quantum circuit
Weights $\boldsymbol{\theta}$	Gate angles
Loss	Cost C
SGD step	Gradient step
Forward pass	Circuit evaluation

Two bottlenecks:

1. Circuit overhead: $O(L^2 n)$ circuits/step
2. Ansatz expressibility

Our work attacks **bottleneck 1**.

The Pauli Overhead Problem

Any $2^n \times 2^n$ matrix decomposes as

$$A = \sum_{P \in \{I,X,Y,Z\}^{\otimes n}} c_P\, P, \quad c_P = \frac{1}{2^n}\,\mathrm{Tr}(A\,P).$$

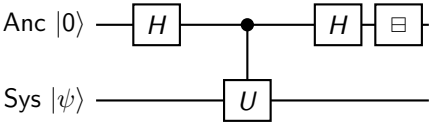
VQLS evaluates $\langle b|A|\psi\rangle$ via *Hadamard tests* — one circuit per Pauli pair.

Cost per gradient step:

$$\#\text{circuits} = L^2(n+1),$$

where L = number of nonzero Pauli strings.

- $N = 4$, all Paulis: $16^2 \cdot 3 = \mathbf{768}$ circuits/step
- $N = 8$, all Paulis: $64^2 \cdot 4 = \mathbf{16{,}384}$ circuits/step



Hadamard test (one per Pauli term)

Bottom line

$L \propto N^2$ terms $\Rightarrow L^2 \propto N^4$ circuits per step.
Halving L gives a $4\times$ speedup.

The Basis-Change Identity

Key observation

The collocation matrix in *any* polynomial basis $\{\phi_j\}$ is

$$A_C = A_{\text{mono}}\, C^\top,$$

where C_{jk} = coefficient of t^k in ϕ_j. The physical solution u is **independent of the basis choice.**

Consequence:

- $C = I$ gives the monomial basis (Case A) with up to 4^n Pauli strings
- Any invertible C gives a valid discretization
- The Pauli coefficients of A_C are *linear* in the entries of C

LP for Pauli minimization

$$\min_{C,\mathbf{t}} \sum_P t_P \quad \text{s.t.} \quad |c_P(C)| \le t_P, \quad \det C \neq 0$$

Each c_P is linear in C, so this is a **linear**

Global LP vs. Monic LP

Global LP — optimize all N^2 entries of C:

- Always finds $A^* = I_N$
- Only 1 Pauli string ($I^{\otimes n}$)
- Problem: $I_N |x\rangle = |b\rangle$ means $|x\rangle = |b\rangle$ — trivially classical, **no quantum advantage**

Monic LP — restrict C to lower-triangular, unit diagonal:

$$C = \begin{pmatrix} 1 & & & \\ & 1 & & \\ & * & 1 & \\ & * & * & 1 \end{pmatrix}$$

- Only $\frac{N(N-1)}{2}$ free variables
- $A^* \neq I_N$ — system stays non-trivial
- No extra normalization constraint needed
- Moderate but genuine Pauli reduction

The monic LP is the principled middle ground

Results: Pauli String Counts

Experiment	N	Case A	Monic LP	Global LP
$u'' = 0$, $[0, 1]$	4	16	**6**	1
$u'' + u' + u = f$, $[-1, 1]$	4	16	**10**	1
$u'' + u' + u = f$, $[-1, 1]$ (8-point)	8	64	**36**	1

Hadamard tests per gradient step:

	Case A	Monic LP	Global LP
$N = 4$	768	108	3
$N = 8$	16 384	5 184	4

Key takeaway

Monic LP gives a **7**$\times$ speedup for $N = 4$ and **3.2**$\times$ for $N = 8$ in circuit evaluations, without collapsing the quantum problem to a classical one.

Timing Comparison Across System Sizes

Wall-clock time per VQLS gradient step on a CPU simulator (PennyLane, `lightning.qubit`).

Size	n	Case A (Monomial)		Monic LP		Global LP (I_N)	
		Paulis	s/step	Paulis	s/step	Paulis	s/step
4×4	2	16	17.2	6	1.3	1	0.05
8×8	3	64	$\sim$240	36	$\sim$75	1	0.11
16×16	4	256	$>$3000 *	112	$\sim$920 *	1	$\sim$0.3

*Extrapolated from $N^4 \log N$ scaling; Case A at $N = 16$ is not feasible on CPU.

Scaling: Cost $\sim L^2 n \sim N^4 \log N$
Every doubling of N increases Case A cost by $\approx 16\times$.

LP speedup grows with problem size

$N = 4$: $\approx 13\times$ $N = 8$: $\approx 3.2\times$
$N = 16$: $\approx 3.3\times$ (projected)

Convergence and Ansatz Limitation

Experiment 1 ($u'' = 0$, Global LP):

- Case A: cost 0.183 after 20 steps ($\approx$344 s). Stuck.
- Global LP: converges to $\approx$0 in **13 steps** (0.7 s)

Experiment 2 ($u'' + u' + u = f$, Global LP):

- Case A: stuck at 0.187 after 20 steps
- Global LP: drops to 0.025 in **4 steps** (0.25 s)
- Plateau at 0.025 = ansatz expressibility limit

Two independent bottlenecks

1. **Circuit overhead:** $O(L^2 n)$ Hadamard tests/step
   Fixed by LP preprocessing
2. **Ansatz expressibility:** circuit must represent $|x\rangle$
   Not affected by Pauli count — requires deeper ansatz

Both LP formulations share the same ansatz plateau — the overhead reduction is the sole benefit.

Conclusion

What we showed:

1. The collocation matrix is freely adjustable via basis change without altering the solution — enabling LP-based Pauli minimization
2. **Global LP** always finds $A^* = I_N$ (1 Pauli), but makes the problem classically trivial
3. **Monic LP** finds a non-trivial $A^* \neq I_N$ with substantially fewer Paulis: $7\times$ speedup at $N = 4$, growing with N
4. Two independent bottlenecks govern VQLS: circuit overhead (cured by LP) and ansatz depth (separate issue)
5. In 1D, reducing Paulis removes quantum

Practical recommendation

For VQLS implementations on near-term hardware: apply the **monic LP preprocessing** before solving. Costs milliseconds of classical LP time; saves orders-of-magnitude in quantum circuit evaluations.

Future work

- 2D Laplacian: joint basis-change LP over shared $B^\top$
- Chebyshev/Legendre node optimization
- Deeper ansatz to overcome

Thank you!

Questions?

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